4 Simplified Neuronal Models

4.1 The Van der Pol Equations

. . . and perhaps also heart-beats.
—B. Van der Pol, 1926 [VdP26]

In the 1920s, Balthasar Van der Pol was working at the Philips Light Company in the Netherlands. He has an interesting background. Trained as a physicist, his mentors were J. J. Thompson, who discovered the electron, and H. A. Lorentz, whose theoretical work helped lay the mathematical foundation for special relativity.

At Phillips, Van der Pol was studying vacuum tubes—triodes. These had an internal filament surrounded by a grid, and a plate on the outside of this inner portion. One could charge the inner filament with a negative charge, and with a potential difference with respect to the plate, send electrons from one end to the other. One could either block or amplify such currents with the potential on the grid that lay in between. The arrangement is shown in figure 4.1.

In 1926, in the *London, Edinburgh, Dublin Philosophical Magazine and Journal of Science*, Van der Pol published a paper entitled *On Relaxation Oscillations* [VdP26]. The first thing he did was rather simple. He took the equation of a damped harmonic oscillator

$$\ddot{x} + k\dot{x} + x = 0$$

where $\dot{x}$ represents differential with respect to time (differentiating twice as $\ddot{x}$). He then changed the damping term k to give

$$\ddot{x} + \varepsilon(x^2 - 1)\dot{x} + x = 0 \tag{4.1}$$

This simple nonlinearity, x^2, introduced enormous complexity to the dynamics. Epsilon, ε, makes the damping term large or small.

Van der Pol, for small ε, describes limit cycles in this equation. In figure 4.2A he plots voltage versus the derivative of voltage. Starting at a point near the middle the trajectory spirals out and approaches a circle, and from a point further out on the diagram, spirals in

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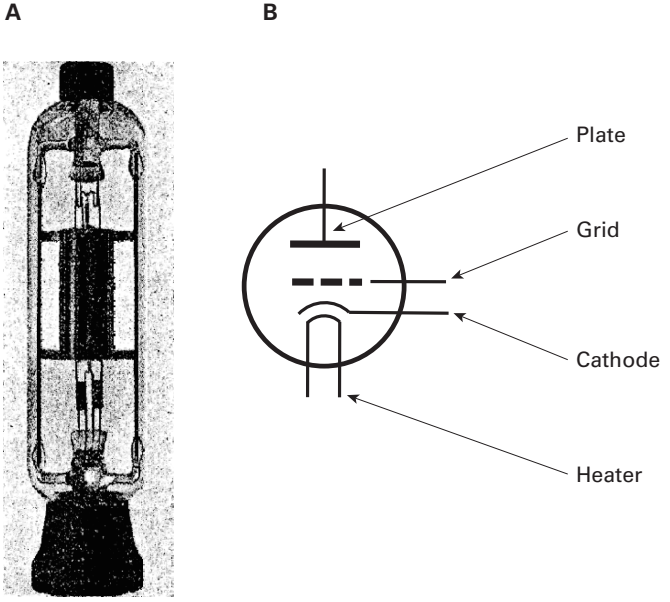


Figure 4.1
The triode. Panel A is a photograph [Kow21] of the first triode used in an electrosurgery circuit [GR03], and B shows its schematic. The grid in the middle, between the filament and the plate at the ends, controls the flow of electrons. Image in A reproduced from [Kow21] with permission.

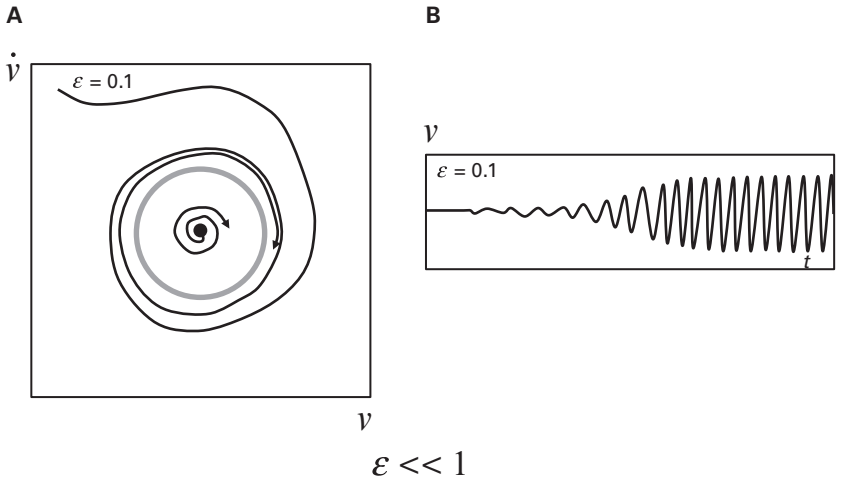


Figure 4.2
Limit cycle in the phase plane V versus $\dot{V}$ (A), and time series of voltage for $\varepsilon \ll 1$ (B). Modified from [VdP26] with permission.

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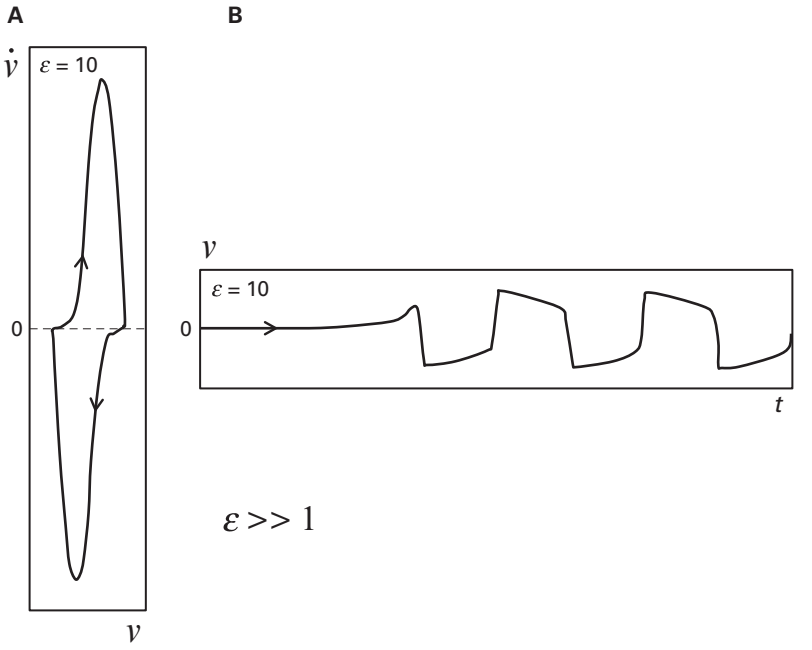


Figure 4.3
Phase plane V versus $\dot{V}$ (A), and time series of voltage for $\varepsilon \gg 1$ (B). Modified from [VdP26] with permission.

toward this same circular set. This is a stable limit cycle. Linear systems do not give stable limit cycles—one sees families of cycles parameterized by initial conditions that are not robust to perturbation. The limit cycle doesn't depend on the initial conditions; you settle down to it starting from an infinite number of potential initial conditions. Limit cycles are stable to perturbation. These are the properties that you want if you are building living systems that need to oscillate robustly. You want your heartbeat to settle down to something fairly similar to the setpoint frequency of heartbeat you want for your heart at rest, regardless of what you did to it this morning. Similarly, you would like your respiratory system to have a very deep limit cycle stable oscillator, so that you tend to breathe in a very cyclic manner.[1] *Life depends on limit cycles*.

If one plots voltage over time for small values of ε, one tends to see very stable sine wave oscillations, as in figure 4.2B—one frequency. Van der Pol, however, was interested in what happens when the damping coefficient is large, $\varepsilon \gg 1$. Multiple frequencies emerge. What was a limit cycle now looks strange and distorted in figure 4.3. Note in the time series plotted in figure 4.3 that the system appears to asymptotically approach one voltage, then

1. Why circadian rhythms from flies to humans do not cycle about a stable limit cycle is an open evolutionary question [PHK02].

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takes a sudden rapid jump toward another voltage, and it flips and flops back and forth. It is bistable. Van der Pol termed these kinds of multistable systems *relaxation oscillators*. It is perhaps an unfortunate term.[2]

A key feature of the dynamics where ε is much different from 1 is shown in equation (4.1), where the dynamics have two different time scales (we will show this explicitly later in this chapter). In the time series plot of the relaxation oscillator in figure 4.3B, therefore, one could make the dwell time along the plateaus shorter (similar to neuronal action potentials), or longer (similar to heart cell action potentials), by tweaking the parameter ε. Van der Pol ends his 1926 paper speculating that the types of dynamics he has just described were similar to cardiac cycles—*and perhaps also heartbeats*.

4.2 Frequency Demultiplication

One might ask what would happen to the solutions of equations such as (4.1) if they were driven by, for instance, a sine wave. In 1927, Van der Pol and Van der Mark published a description of such driven dynamics in *Nature* [vdPVdM27]. What they found was that the frequencies of relaxation oscillators could produce a strange effect when driven by a given frequency, ω: They would *frequency demultiply*, that is, divide.

Using the simple circuit shown in figure 4.4, Van der Pol and Van der Mark drove it with $Fsin(\omega t)$. By adjusting the capacitor, C, they found frequency division up to 1/200. As they increased the strength of the capacitor, they observed what sounded like noise before some of the transitions (figure 4.4). This driven nonlinear system was not generating random noise. They were observing chaotic dynamics—highly irregular deterministic dynamics with many different frequencies. We will examine chaotic behavior of nonlinear systems in more detail in the control frameworks discussed later. Such dynamics are a challenge for any nonlinear control system.[3]

4.3 Bonhoeffer and the Passivation of Iron

It simply is not enough to know that nerve fibers conduct . . . we need a much more basic and detailed understanding of these stimulus response systems. . . .
—M. Delbrück, 1969 Nobel Address

2. The *relaxation* part was because if you looked at RC circuits, the same time constant obtained from multiplying the effective resistance and capacitance of these circuits would give you the time constant. The reciprocal of the time constant would give you the period of these events, so it was like a relaxation circuit, or rather the relaxation of the RC circuit, and he called it a relaxation oscillator. There appears unfortunately to be some flexibility in the application of this phrase in the literature—but this is actually what it applies to.

3. At present, we leave out of this discussion whether neuronal systems generate chaotic dynamics [SJD+94]. Certainly they do not in an obvious sense. But if we are going to attempt to track nonlinear systems, we should be able to track them when they are at their most erratic. And the methods to be introduced later in this book can impressively track, and control, nonlinear systems even when they are chaotic.

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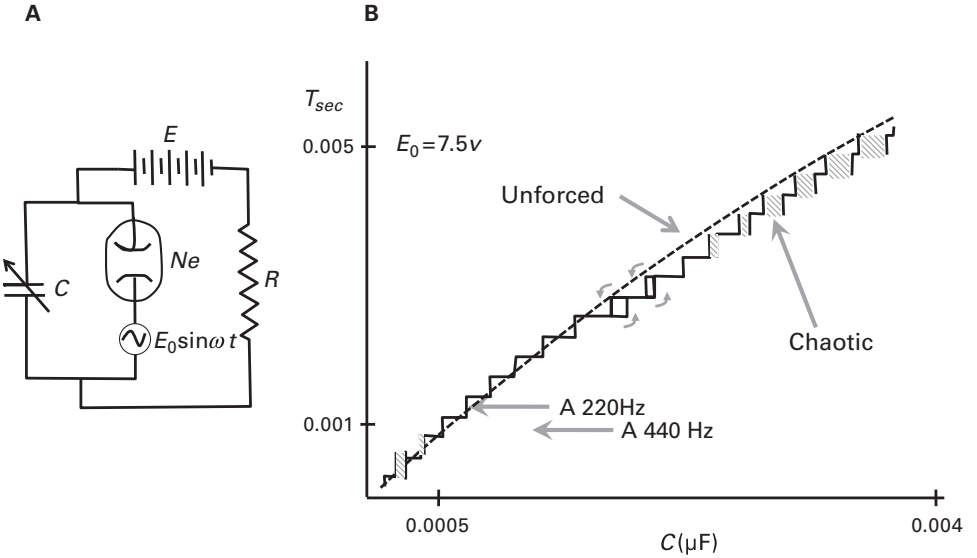


Figure 4.4
Driving a simple relaxation oscillator circuit would produce frequency demultiplication as well as chaotic dynamics. Modified from [vdPVdM27] with permission.

In 1947, Max Delbrück translated an article by Karl Bonhoeffer that discussed work performed during the war years in Germany [Bon48]. Delbrück had been studying lipid bilayers and was quite interested in nerve conduction.

Bonhoeffer would take a metal wire and have a reference electrode with a salt bridge and opposing electrode, as shown in figure 4.5A. Iron does some strange things. One of the reasons stainless steel does not rust is that chromium is added to it, forming an oxide on the surface of iron. Chromium oxide protects the iron (passivation) from corrosion. Nitric acid is used to remove iron atoms from the surface of the steel, and this quickly promotes formation of the chromium oxide. The oxide layer is visible when you watch this happen, and disappears during recovery.

The chemical activation of iron in nitric acid in response to an electrical pulse has a threshold and propagates as a wave; it seems to have an all-or-none response. A refractory period occurs, during which you have to wait for it to get ready to be activated again. It seems to accommodate, which means the reaction does not start if you gradually turn up the current pulse that you are using to activate passivation. And the reaction can oscillate.

Bonhoeffer likens this to ignition of an explosive compound. Figure 4.5B is heuristic. There is a stable point. Warm it a bit and it cools back down to the stable point. Warm it beyond a certain point, and it will actually begin to get hotter and hotter even if you take the

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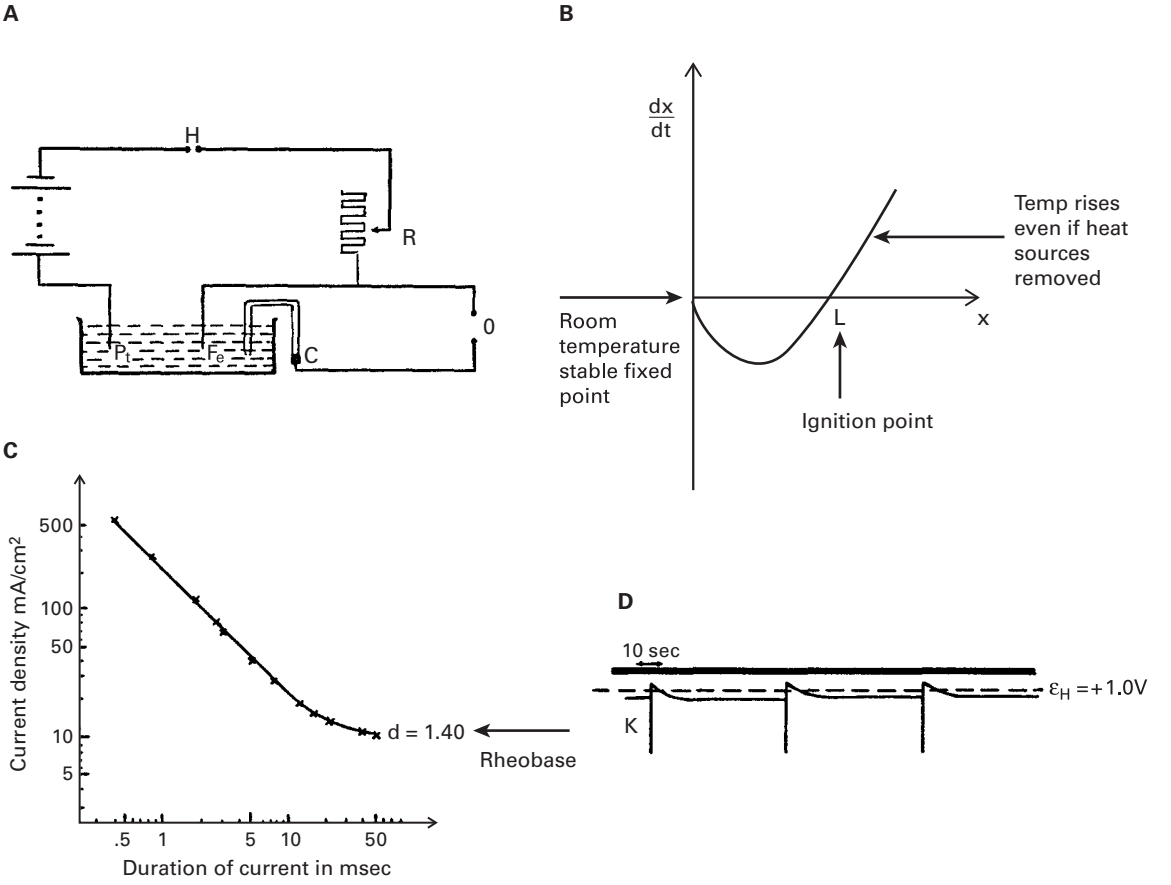


Figure 4.5
A Panel shows the chemical test apparatus for the electrical stimulation of passivation of iron. In B is a heuristic sketch for the nonlinear regenerative process for the ignition point of an explosion. The current-duration curve for activation of iron is shown in C, along with the equivalent of rheobase for analogous neuronal membrane properties. In D, the periodic relaxation oscillations of iron at a constant electrical current resemble features of the activation of neural tissue. Modified from [Bon48] with permission.

heat source away. Similar to this inflammation point of an explosive mixture, there appears to be an ignition point in the activation of iron.

There is also a point where, if you provide a low enough current, nothing happens at all (figure 4.5C). This is similar to a classic plot that you will see in the neuroscience literature, which is current versus duration of pulse required to activate a neuronal membrane. If you give a very short pulse, you need a big current to activate the system. As the pulse gets longer, you actually need less and less current until you reach a place where even an infinitely long pulse will not activate the process. This relationship holds in nerves

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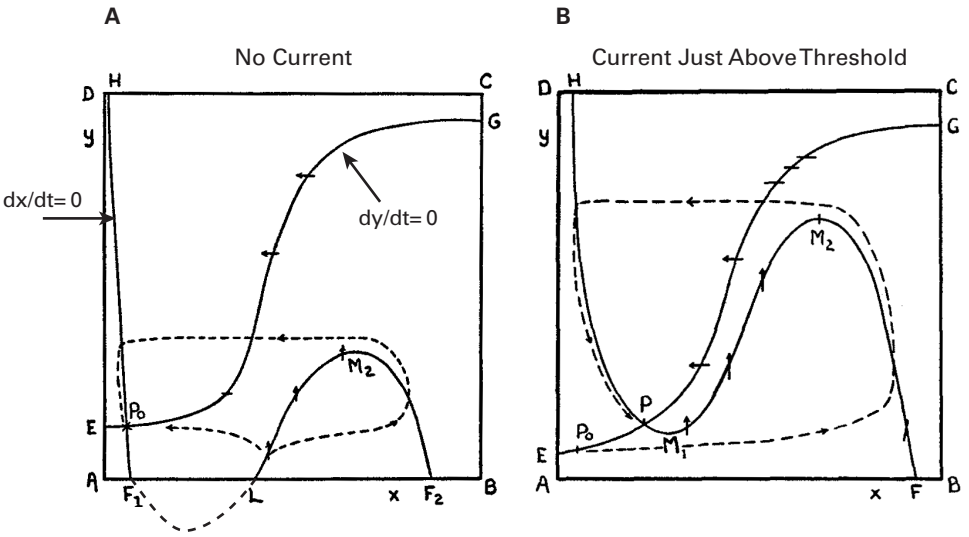


Figure 4.6
Chemical kinetics of the passivation of iron with nullclines plotted without current (A) and with current just above threshold (B).

and in iron bars. Bonhoeffer knew this and called this rheobase, after the neural phenomenon.[4]

Figure 4.5D shows some of the oscillations that Bonhoffer recorded while electrically stimulating at a constant level. He recorded spike-like currents that spontaneously relaxed. It sort of looks like a heartbeat or a spiking reaction of a neural system.

Bonhoeffer knew of Van der Pol's work, and recognized these events as relaxation oscillations. He formulated the dynamics of iron passivation as chemical reactions, assuming that there would be an activation variable, x, and a refractory variable, y. Bonhoeffer recognized that with analogy to electrical circuits, there needed to be a *negative resistance* to counter the tendency of such resistive circuits to dampen out oscillations (recall the discussion of negative resistance in Hodgkin-Huxley dynamics in chapter 3). His intuition was sufficiently strong that he could qualitatively sketch the characteristic nullclines[5] from a generic set of differential equations, which would account for the characteristics of the passivation of iron (figure 4.6). His qualitative picture of such nullclines provide us with essential intuition regarding neuronal dynamics not only in this chapter, but in our later

4. The related term is chronaxie—the time duration corresponding to twice rheobase. It is a characteristic time for a pulse to activate a neural fiber.

5. Nullclines are plots where the values of the rate of change of a variable are equal to zero. The intersection between nullclines points to solutions of differential equations where nothing is changing. We will repeatedly examine nullcline plots throughout this book.

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discussion of Parkinson's disease dynamics (chapter 10). Without current, the nullclines have only one intersection, as shown in figure 4.6A. Just above threshold current, one nullcline rises and a limit cycle emerges in figure 4.6B. The separation of time scales produces a relaxation oscillation similar to those in figure 4.3 and to be seen shortly when we modify these equations in the next section.

4.4 Fitzhugh and Neural Dynamics

The possibility of representing excitable systems by a generalization of the van der Pol equation was suggested to the author by Dr. K. S. Cole.
—R. Fitzhugh [Fit61]

Richard Fitzhugh was one of the first to pioneer the phase plane analysis of neuronal models with the view to understand their dynamical properties [Izh07]. He began working on a better understanding of the Hodgkin-Huxley equations, and was the first to recognize that the Hodgkin-Huxley model was *one member of a large class of nonlinear systems showing excitable and oscillatory behavior*—including both Van der Pol's triodes and Bonhoeffer's passivation of iron [Fit61]. Fitzhugh recognized that one of the values of *state space* diagrams is that each point on the diagram represented a particular state of the system. Different regions of the diagram corresponded to different physiological states of the nerve membrane. He further recognized that the study of the few *fixed points*, stable or unstable solutions that were in the phase plane, were the skeleton of the dynamics.[6] Understand the fixed points, and you understand a great deal about the qualitative nature of the system.[7]

Starting with Van der Pol's equation (4.1),

$$\ddot{x} + \varepsilon(x^2 - 1)\dot{x} + x = 0$$

Fitzhugh introduced a Liénard transformation,

$$\dot{x} = \varepsilon\left(y + x - \frac{x^3}{3}\right)$$

which lets you write (see exercise 4.1) the two first-order differential equations:

$$\begin{aligned} \dot{x} &= \varepsilon\left(y + x - \frac{x^3}{3}\right) \\ \dot{y} &= -x/\varepsilon \end{aligned} \tag{4.2}$$

6. Large numbers of unstable fixed point solutions are characteristic of chaotic systems. The intuition that they form the *skeleton* of such dynamics was first recognized by Predrag Cvitanovic [Cvi88].

7. In a book about state estimation, these state space diagrams provide an invaluable way to visualize state geometrically, and also to construct geometrical versions of control laws. We are generally limited in their use, of course, to the cases where model reduction permits us to plot in the two or three dimensions that we can visualize.

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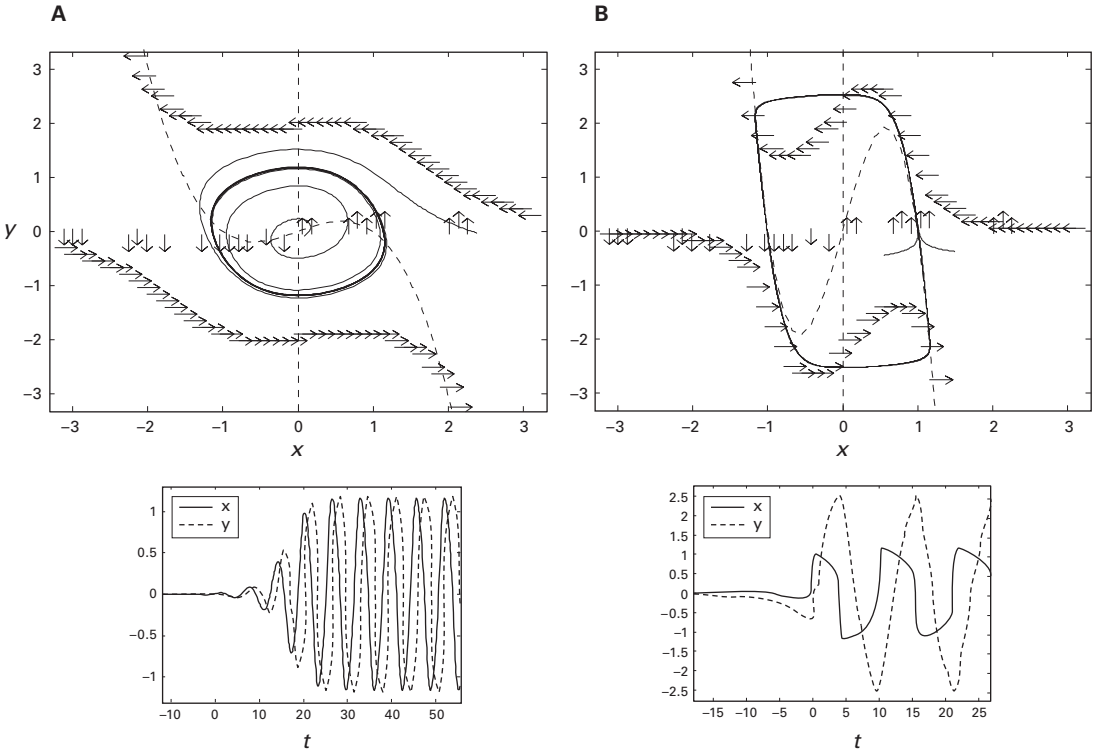


Figure 4.7
Nullclines (dashed lines) for the transformed Van der Pol equations (4.1) are $x = 0$ and $y = \varepsilon(x - x^3)$. In A are shown the results of two solutions with $\varepsilon = 0.5$, and below a set of time series for one of the solutions for x and y. In B are shown two solutions with $\varepsilon = 5.0$, and below a set of time series for one of the solutions. Prepared with pplane7 by John Polking, with permission.

The term $(x - \frac{x^3}{3})$ is a linear minus a cubic function, and this leads to an "N"-shaped function if plotted.

Setting the derivatives in equation (4.2) to zero gives us the nullclines plotted in figure 4.7. In panel A, the limit cycle is calculated for $\varepsilon = 0.5$, while in panel B, the relaxation oscillation is shown for $\varepsilon = 5.0$.

Notice that this particular transformation makes the relative time constants very clear; the rate of change in x, if ε is large, is fast; the rate of change of y, if ε is large, is slow. This shows the separation in time scales very clearly.

Fitzhugh next added a current, I, to the fast x variable to provide stimulation. He added two adjustable parameters, an a and b, putting y into the second equation so that y is represented in both. The slower variable y will become a *recovery* variable—it takes the role of inactivation and potassium in the Hodgkin-Huxley equations (chapter 12).

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$$\begin{aligned}\dot{x} &= \varepsilon\left(y + x - x^3/3 + I\right)\\ \dot{y} &= -(x - a + by)/\varepsilon\end{aligned} \tag{4.3}$$

When x goes up, y will decrease at a slower rate (an effective lag), and as y subsequently increases this causes x to decrease (see figure 4.7B time series).[8] Critical to adding by to the second equation in (4.3) is that he now had an adjustable slope on the $\dot{y} = 0$ nullcline, rather than the straight up and down line shown in figure 4.7.

He also assumed the following bounds on the constants:

$$\begin{aligned}1 - 2b/3 < a < 1\\ 0 < b < 1\\ b < c^2\end{aligned} \tag{4.4}$$

Prescribe the current, I, and you had the analog of the *current clamp* of [HHK52].

The nullclines are now at

$$y = x^3/3 - x - I$$

$$y = -x/b + a/b$$

Let's examine the $\dot{x} = 0$ cubic nullcline first. If I were zero, then the $(x^3 - x)$ term would be "N"-shaped through (0, 0). Conditions (4.4) require that at $I = 0$, the nullclines intersect at only one stable point—the resting state. Adding or subtracting I from the $\dot{x} = 0$ nullcline will raise or lower it. The $\dot{y} = 0$ nullcline is a straight line with an adjustable slope. Adding I will determine the stability of the intersection of these nullclines. Fitzhugh's stability diagram, labeling the different parts of the phase space with the nature of the system, is shown in figure 4.8.[9] If one applies an instantaneous pulse of current I to the resting point P, then the system solution instantly shifts along the dotted line to the left. There is labeled a *separatrix*—literally separating the phase space into two regions where the trajectories are different. If the pulse of current pushes the state across this separatrix, then a *regenerative* spike develops as a large excursion in the voltage x.

It is immediately apparent from figure 4.8 that there is no voltage x for which there is a single voltage threshold. Because the $\dot{y} = 0$ nullcline is tilted, unlike the special case of the Van der Pol equations in figure 4.6, there is a family of values of voltage that forms the threshold. Threshold for such systems is at least a line, or *manifold*. But thresholds for many neuronal models are often thicker than lines—they are *threshold sets*. Within the threshold

8. Fitzhugh, in a humble way, termed these equations the *Bonhoeffer–Van der Pol* model.

9. It is of interest that this diagram was produced by Fitzhugh using an analog computer. His term *no man's land* was a euphemistic reference to the difficulties of reproducing trajectories that reliably crossed into certain regions of the state space using the analog computation available.

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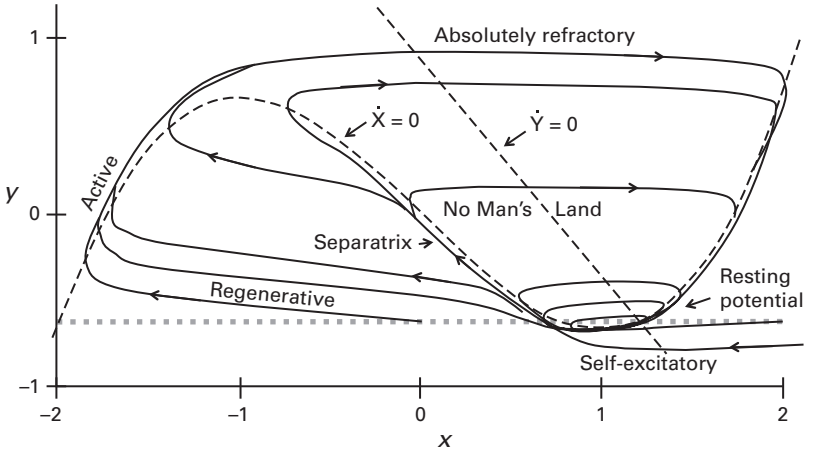


Figure 4.8
Phase plane diagram of the Fitzhugh modification of the Van der Pol equations. The nullclines are placed assuming that the current $I = 0$. Reproduced from [Fit61] with permission.

sets, partial action potentials can be seen, as with the Hodgkin-Huxley equations (see exercise 3.2). For both these equations and the Hodgkin-Huxley equations, *all intermediates between all and none responses can be obtained by adjusting the stimulus with accuracy*. A superb in-depth discussion of the different types of threshold can be found in [Izh07].

By applying a constant negative current I, the $\dot{x} = 0$ nullcline moves up, and this shifts the relationship of the nullcline intersection and the threshold separatrix. In figure 4.9A, the point at which constant current moves the intersection point P just to the position of the new separatrix is the geometrical analog of *rheobase* (see figure 4.4C). If positive constant current is applied, the $\dot{x} = 0$ nullcline moves downward, and on release of this current the system state might be below the separatrix, resulting in anode break excitation (figure 4.9B).

And finally, Fitzhugh shows that a sufficiently large negative current raises the $\dot{x} = 0$ nullcline sufficiently high to make the intersection unstable, and a stable limit cycle appears as in figure 4.10. An infinite series of periodic spikes ensues.

Fitzhugh noted in an earlier paper [Fit60] that the time courses of the Hodgkin-Huxley variables n and h were nearly inverses of each other—projected on the $n - h$ plane the trajectory followed close to a straight line. He experimented with reducing the four-variable Hodgkin-Huxley equations to a two-dimensional form with the transformations $w = (n - h)/2$, and $u = V - 36m$. Such reductions showed qualitative similarity with his modified Van der Pol equations (4.3). But the two-dimensional phase space of (w, u) were projections of four variables, and not a true phase space (where each point represents a single state of the system). Improving this will need to await John Rinzel's later work (discussed below).

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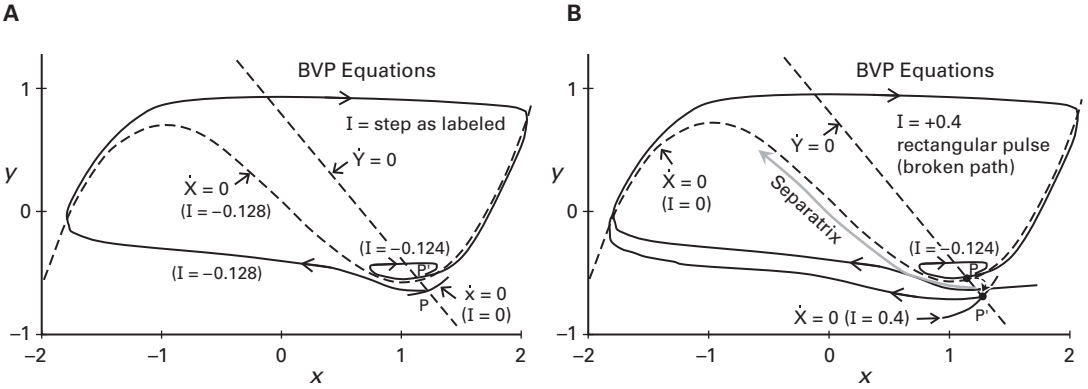


Figure 4.9
Effect of constant current moving the height of the $\dot{x} = 0$ nullcline. Rheobase is illustrated in A, and anode break excitation in B. Modified from [Fit61] with permission.

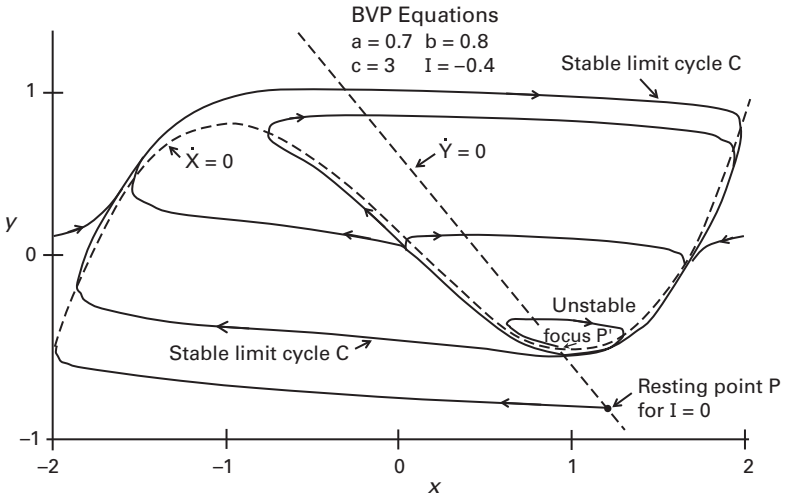


Figure 4.10
Effect of constant current moving the height of the $\dot{x} = 0$ nullcline sufficiently high so that the nullcline intersection becomes unstable and a stable limit cycle is produced. Reproduced from [Fit61] with permission.

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Last, Fitzhugh noticed that neither the Hodgkin-Huxley equations nor his modified Van der Pol equations, replicated what real squid axons did in response to large constant current pulses. Real axons got tired and stopped firing—the models never showed such behavior. Fitzhugh recognized that work by Frankenhaeuser and Hodgkin [FH56] demonstrated that, with each action potential, there accumulated a small amount of potassium outside of the squid axon. This would serve to decrease the potassium equilibrium potential gradually over time, and would eventually lead to axonal unresponsiveness. Later, we will introduce a variable potassium into the Hodgkin-Huxley equations in order to be able to reconstruct and track such accumulating potassium. We will further show how the slow dynamics of potassium shifting out of and into cells is a critical piece of seizure dynamics, and how, in a control theoretic framework, not accounting for potassium leads to tracking failure in such networks (chapter 12).

The Bonhoeffer–Van der Pol model is not intended to be an accurate quantitative model of the axon, in the sense of reproducing the shape of experimental curves; it is meant rather to exhibit as clearly as possible those basic dynamic interrelationships between the variables of state which are responsible for the properties of threshold, refractoriness, and finite and infinite trains of impulses. (R. Fitzhugh, 1961 [Fit61])

4.5 Nagumo's Electrical Circuit

Jin-Ichi Nagumo and colleagues [NAY62] were interested in replicating the lack of attenuation and distortion characteristic of neural signal propagation in an electrical communication channel. Using Fitzhugh's formulation of equations (4.3), they used Kirchhoff's laws and the new tunnel diodes[10] to implement Fitzhugh's equations using the circuit shown in figure 4.11A. By chaining these elemental circuits, as in figure 4.11B and C, they could replicate the propagating action potentials shown in figure 4.11D and E. Note, in particular, that these propagating pulses were stable. Starting with a narrow pulse, it widens to an asymptotic width (figure 4.11D) that is the same as what a wide initial pulse contracts to (figure 4.11E).[11]

Fitzhugh's modification of the Van der Pol equation in (4.3) are now most commonly referred to as the *Fitzhugh-Nagumo* equations.

4.6 Rinzel's Reduction

John Rinzel overlapped for many years with Richard Fitzhugh at the Mathematical Research Branch at the U.S. National Institutes of Health. In 1985, Rinzel offered an alternative reduction of the Hodgkin-Huxley equations to the one Fitzhugh had suggested earlier.

10. The tunnel diode was invented in 1958 by Leo Esaki, who in 1973 received the Nobel Prize in Physics for discovering the electron tunneling effect used in these devices.

11. Competing with Fitzhugh for humility in describing their own results, they write that it *behaves similarly to the real axon but the representation is gross.*

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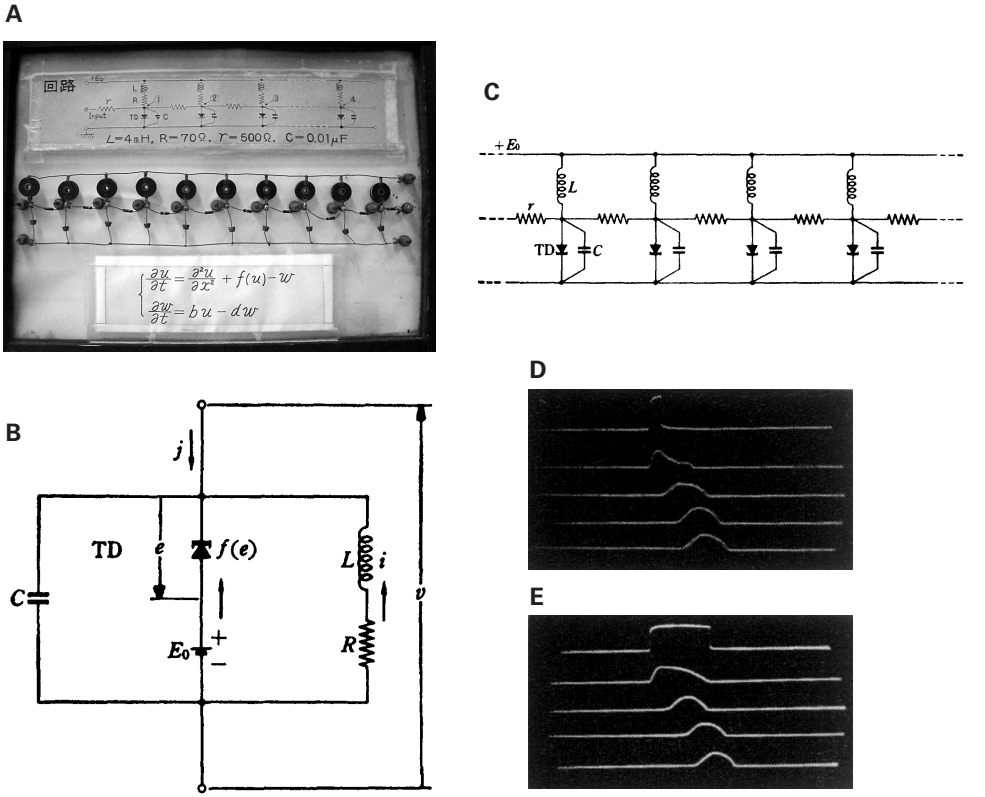


Figure 4.11
Nagumo's circuit is shown in A. Equivalent circuit to equations (4.3) are shown in B, and assembled to replicate a propagating action potential in C. The dynamics of the propagating pulse demonstrate asymptotic stability of the traveling wave as a narrow pulse widens, in D, and a wide pulse narrows, in E, each toward the same shaped action potential wave. B, C, D, and E reproduced from [NAY62] with permission. I am indebted to Professor Kazuyuki Aihara for kindly providing the photograph of Professor Nagumo's original circuit in A.

Taking advantage of the linearity of n versus h, he suggested fitting the line $h + Sn = 1$. The slope, S, was determined by the resting values of n and h as $S = (1 - h_0)/n_0$. This let him define a single recovery variable as $W = S[n + S(1 - h)]/(1 + S^2)$, from which two variables are replaced by one. He further notes that m is the fastest gating variable in Hodgkin and Huxley's equations, so one might as well just assume it is always at its infinite time value, or $m = m_\infty(V)$. Setting $h = (1 - W)$ and $n = W/S$, he develops a true two-variable reduction of the Hodgkin-Huxley equations:

$$I = C_M \frac{dV}{dt} + \bar{g}_K (W/S)^4 (V - V_K) + \bar{g}_{Na} m_\infty^3(V)(1 - W)(V - V_{Na}) + \bar{g}_l (V - V_l)$$
$$\dot{W} = \phi[W_\infty(V) - W]/\tau(V) \tag{4.5}$$

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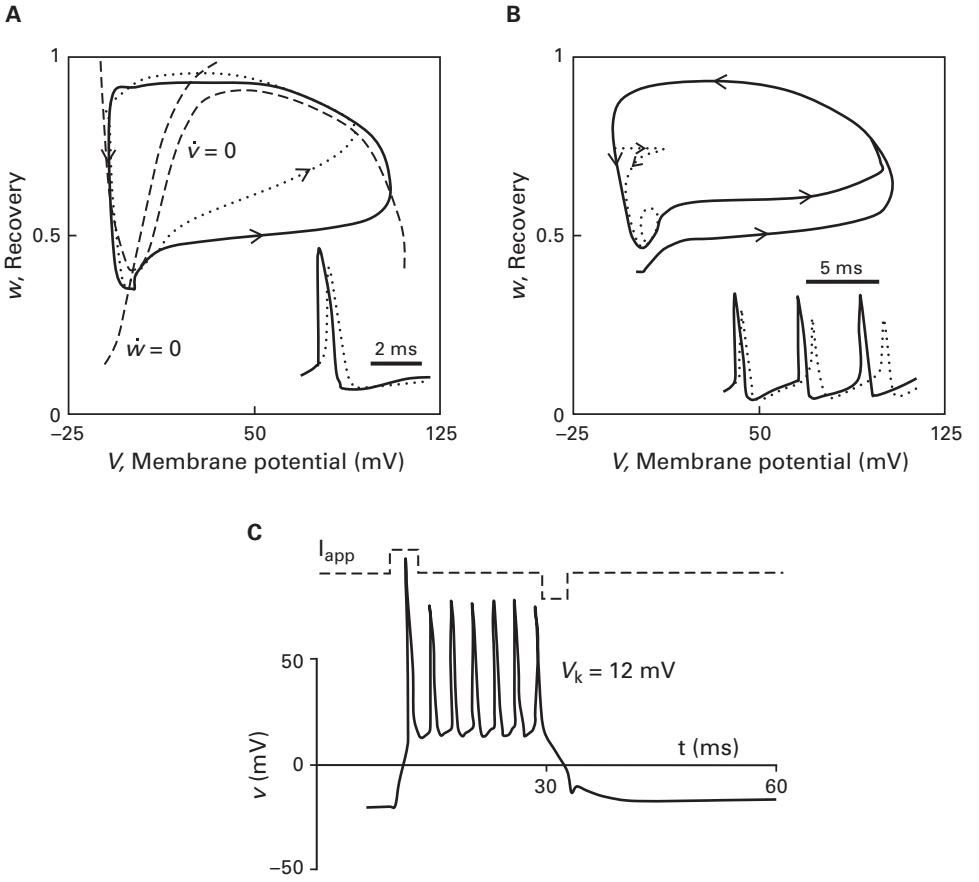


Figure 4.12
Similarities of full and reduced Hodgkin-Huxley equations to brief current pulse, A, and current step in B. Full model results are dotted, reduced Hodgkin-Huxley results are solid. C shows the bistability of reduced Hodgkin-Huxley equations under conditions of elevated potassium. No equivalent for Fitzhugh-Nagumo can replicate C. Reproduced from [Rin85] with permission.

where $W_\infty(V) = S[n_\infty(V) + S(1 - h_\infty(V))]/(1 + S^2)$, and $\tau(V)$ was a compromise between $\tau_h(V)$ and $\tau_n(V)$ as $\tau(V) = 5exp[-(V + 10)^2/55^2] + 1$. ϕ was a temperature correction factor. He further prescribed $h = 0$ if $W > 1$.

The first thing to notice was that Rinzel's reduction of the Hodgkin-Huxley equations gave a very good approximation to the Fitzhugh-Nagumo equations and the full Hodgkin-Huxley model (figure 4.12A and B). But now with explicit ions, such as potassium, one could perform simulations for which the Fitzhugh-Nagumo equations were not applicable. Figure 4.12C shows that, for elevated extracellular potassium, bistability could be shown in

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these reduced equations, similar to the bistability that John Moore had earlier demonstrated experimentally in squid axons [Moo59].

4.7 Simplified Models and Control

In 2004, Henning Voss, Jens Timmer, and Jürgen Kurths [VTK04] applied an unscented Kalman filter to estimate variables and parameters of the Fitzhugh-Nagumo equations. This will be discussed at length in chapter 5. Nevertheless, it is important to recognize that there are many other ways to boil down the dynamics of a neuronal membrane, or the Hodgkin-Huxley equations, to a smaller number of dimensions. To the extent that such models preserve relevant dynamics of a given neuronal system, the computational efficiency gained may introduce substantial benefit. The interested reader is urged to consult [GK02] or [Izh07] for a variety of other useful formulations. But from a philosophical control perspective, we will in a later chapter analyze in some detail the advantages and disadvantages of model fidelity to the biophysics, versus fidelity to the dynamics. The entire issue of *model inadequacy* (chapter 8) lays at the cutting edge of control theory. But we are getting ahead of ourselves. First, we need to apply the unscented Kalman filter to the Fitzhugh-Nagumo equations in the next chapter.

Exercises

4.1. Derive equations (4.2) using (4.1) and the Liénard transformation $\dot{x} = \varepsilon \left(y + x - \frac{x^3}{3} \right)$.

4.2. Write a program to integrate the Fitzhugh-Nagumo equations (4.3). Use an Euler integration scheme. Set values of $a = 0.7$, $b = 0.8$, and $\varepsilon = 3$. I would suggest starting with an integration time of 0.01, and integrating over perhaps 100 units of time. Try applying pulses for the middle half of this time. Use both positive and negative pulses between -1 and $+1$.

- What range of current gives subthreshold responses?
- What range of current produces limit cycles?
- Can you find a range of current that produces anode break excitation?

4.3. Euler integration, as we have been using, calculates the derivative at the endpoint of an interval, and assumes that it is a good approximation to a function throughout that interval. Euler assumes that the derivative of a function, dx/dt, evaluated at x_n and t_n, can be approximated as

$$\frac{x_{n+1} - x_n}{t_{n+1} - t_n} \approx \frac{dx}{dt}(x_n, t_n)$$

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so that the next value being approximated in the integration, x_{n+1}, is equal to

$$x_{n+1} \approx x_n + \Delta t \frac{dx}{dt}(x_n, t_n)$$

where $\Delta t = t_{n+1} - t_n$ as illustrated in figure 4.13. A more accurate way to numerically integrate is to reestimate the derivative along the way from time t_1 to t_2. The second-order Runge-Kutta method (RK2) first evaluates the same product of derivative and time interval as in Euler integration

$$k_1 = \Delta t \ \frac{dx}{dt}(x_n, t_n)$$

which gives an increment in x. But we then take half of this increment, at half of the time interval, and calculate

$$k_2 = \Delta t \ \frac{dx}{dt}\left(x_n + \frac{k_1}{2},\ t_n + \frac{\Delta t}{2}\right)$$

such that

$$x_{n+1} \approx x_n + k_2$$

This intermediate derivative is more accurate than Euler, and the geometry is illustrated in figure 4.13. A detailed authoritative source on numerical integration is Press et al. [Pre07].

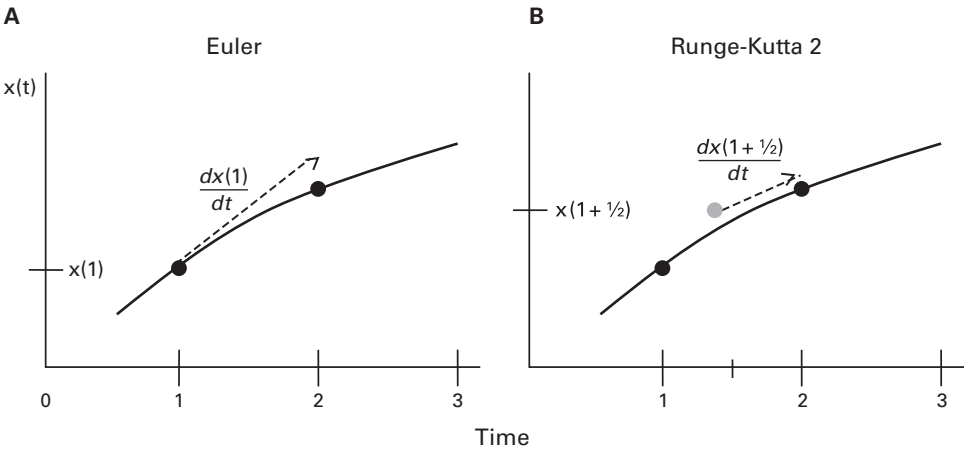


Figure 4.13
Schematic of Euler integration (A) versus Runge-Kutta of second order (B).

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• Write a function that will let you pass parameters and variable values to and just calculate the Fitzhugh-Nagumo equations (derivatives). Such a function will be useful in writing RK2 and higher-order routines. Let the function return a vector of the two Fitzhugh-Nagumo variables, $\dot{x}$ and $\dot{y}$. This vector of derivatives can then be multiplied by the time interval, to generate k_1. By adding $k_1/2$ to the vector of variables, you can reuse this same function to calculate k_2.

• Now modify your program from exercise 4.2 so that it integrates the Fitzhugh-Nagumo equations using RK2 integration.

4.4. Last, the standard for numerical integration in this book is the fourth-order Runge-Kutta (RK4) integrator.[12] This is an extension of the RK2 routine introduced in exercise 4.3.

• You will need an algorithmic routine that calculates the RK4 intermediate quantities, $k_1 \ldots k_4$. One way of structuring this is to use the Fitzhugh-Nagumo derivatives function introduced in exercise 4.3, and call it with the following routine:

```
k1=dT*Fitzhugh-Nagumo_dot(Variables,Parameters);
k2=dT*Fitzhugh-Nagumo_dot(Variables+k1/2,Parameters);
k3=dT*Fitzhugh-Nagumo_dot(Variables+k2/2,Parameters);
k4=dT*Fitzhugh-Nagumo_dot(Variables+k3,Parameters);
xy(:,t)=Variables+k1/6+k2/3+k3/3+k4/6;
```

where dT is the time interval, Fitzhugh-Nagumo_dot is the function that calculates $\dot{x}$ and $\dot{y}$, `Variables` and `Parameters` are column vectors of the variables and parameters in the model, and xy is the full time series of the variables values (each column are the values at one point in time).

12. Press et al. [Pre07] suggest that use of RK4 is for *ploughing fields*. But they also recognize that, properly used, such a workhorse algorithm can serve us well.

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